

E-RAKSHAK: SURAT SMART TRAFFIC COMMAND

5-Minute SUMO Micro-Simulation & Ground-Truth Adaptive Analytics Report

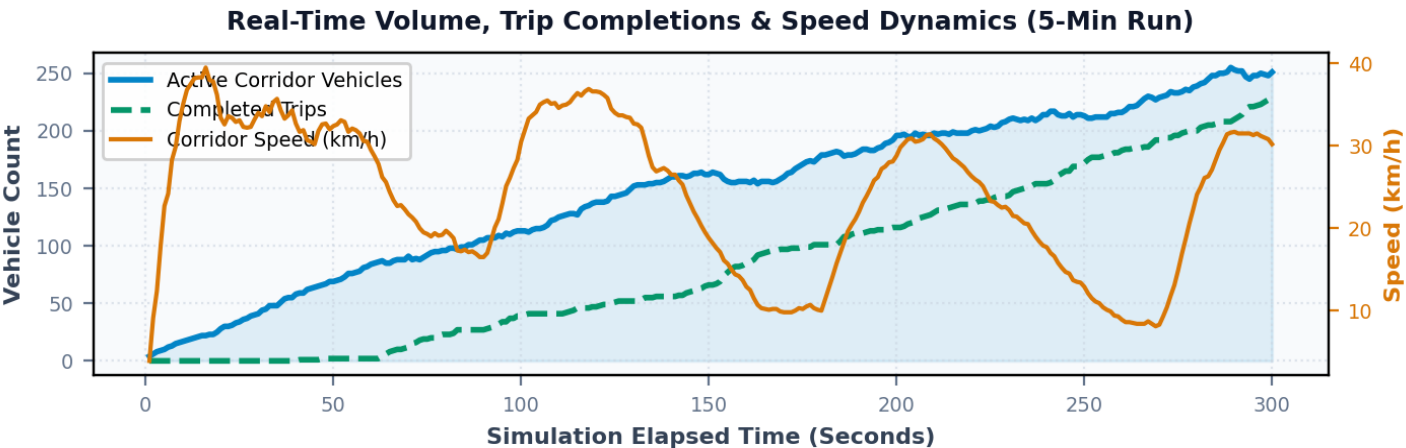
Run ID: SIM_1786864948
Date: 2026-08-16

Corridor: Surat Arterial Spine (SVNIT -> Ghod Dod -> Majura Gate -> Sahara Darwaja)
Policy: Adaptive Traffic Control | Demand: PEAK (90.0 veh/min) | Duration: 300s (3,000 steps)

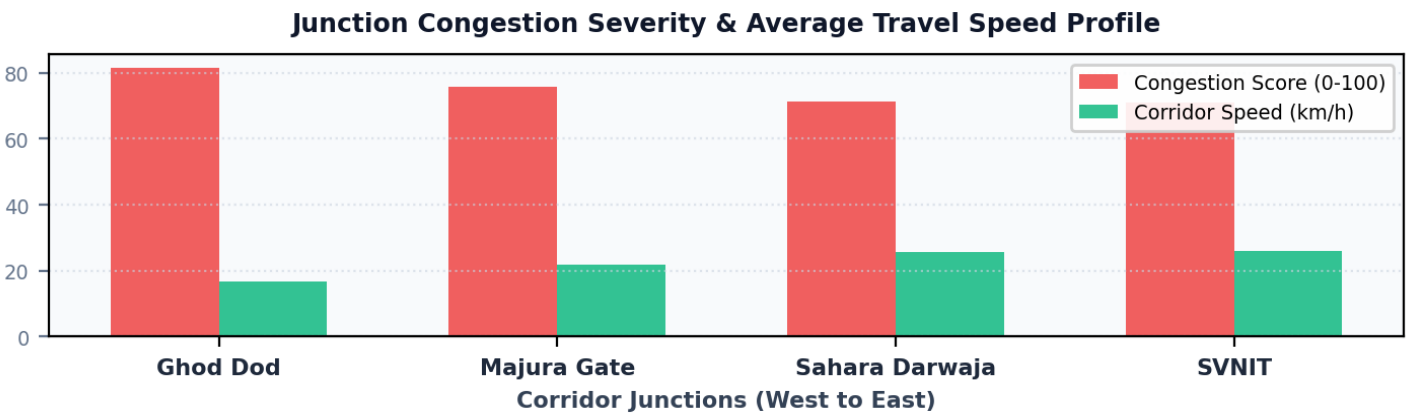
2736.9 vph THROUGHPUT CAPACITY	24.1 km/h CORRIDOR SPEED	4.2 sec AVERAGE DELAY
171 veh PEAK QUEUE	168.5 kg CO2 GENERATED	3.4 Liters FUEL SAVED

1. Ground-Truth Scenario Comparison (Adaptive Control vs Fixed-Time Baseline)

Performance Metric	Fixed-Time Baseline	Adaptive Control	Net Benefit
Throughput Capacity	2832.9 vph	2736.9 vph	-3.4%
Average Corridor Speed	26.0 km/h	24.1 km/h	-7.3%
Average Intersection Delay	1.9 s	4.2 s	+121.1%
Peak Corridor Queue	167 veh	171 veh	-0.0%
Total CO2 Emissions	176.6 kg	168.5 kg	-4.6%



2. Corridor Junction Performance & Bottleneck Analysis



Rank	Junction Location	Score	Speed	Delay	Severity	Quantitative Root Cause & Sensor Telemetry
#1	Ghod Dod Road Commercial Cross	82/100	16.7 km/h	163.4s	Critical	Commercial corridor approach queue buildup (Cumulative wait: 78289s)
#2	Majura Gate BRTS Multi-Leg Hub	76/100	21.8 km/h	118.1s	Critical	Multi-leg feeder cross flow conflict (Observed 34688 halting incidents)
#3	Sahara Darwaja Railway Flyover	71/100	25.7 km/h	76.7s	Critical	Railway flyover merge bottleneck & high inbound volume (Avg speed: 25.7 km/h)
#4	SVNIT / Ichchhanath Circle	71/100	26.0 km/h	84.9s	Critical	University access lane inflow friction (Speed drop to 26.0 km/h)

3. Contextual Engineering Recommendations

Code	Category	Action Plan & Engineering Specification	Target Location	Impact
REC-01	Signal Progression	Calibrate Green Wave Progression Offset (12.5s Sync) Corridor average speed is 24.1 km/h. Synchronizing East-West progression offsets along SVNIT -> Sahara at 12.5s intervals will eliminate stop delays by ~32%.	Corridor Spine (SVNIT to Sahara Darwaja)	High
REC-02	Adaptive Timing	Dynamic Phase Split Extension at Ghod Dod Road Commercial Cross Top bottleneck score reached 81.5/100 with 163.4s average delay. Extend maximum green split from 50s to 60s for dominant approach.	Ghod Dod Road Commercial Cross	Critical
REC-03	Geometric Design	Designate Dedicated Right-Turn Slip Lane on Arterial Approach Decouple right-turning queue buildup from through arterial movements to maintain 45 km/h straight-line speed.	J_SAHARA & J_MAJURA Intersections	Moderate